

ABHINAV KAUSHIK

Senior Data Scientist

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SUMMARY

Senior Data Scientist with 8+ years building ranking, retrieval and recommendation systems and taking them from research to production, with an MSc in Machine Learning and AI from Liverpool John Moores University. Works across the recommender stack: embedding-based candidate generation and vector search, learning-to-rank re-rankers, gradient boosting with LightGBM and XGBoost, and re-rankers shipped into a live recommendation system against commercial targets. Strong Python, SQL and PySpark, with production delivery through Docker, Airflow, MLflow and AWS, plus offline evaluation pipelines and metric design that decide what ships. Drives technical direction across candidate generation, ranking features, logging and evaluation, turning ambiguous product problems into measurable objectives with product, engineering and analytics stakeholders, and uses Claude Code and Cursor to move faster. Looking for opportunities in Berlin and willing to relocate. Currently building ThinkTree.AI, a non-profit AI-powered learning platform designed to support students with ADHD and context-switching challenges.

PROFESSIONAL EXPERIENCE

Senior Data Scientist | United Health Group | Bengaluru, India Nov 2025 – Present

- Fine-tuned a cross-encoder re-ranker in PyTorch on positive and negative pairs over a dense bi-encoder retrieval stage for AARP survey data, improving retrieval accuracy – a learning-to-rank layer on top of embedding-based candidate generation.
- Engineered a production multi-agent analytics system in LangGraph and PySpark for AARP medical plans, answering business questions with multi-turn reasoning and live charts, cutting manual analysis by 98%.
- Designed automated offline evaluation and observability pipelines with LLM evaluators, tracing and failure detection, reducing human-in-the-loop review by 43% and making model changes measurable before release.
- Shipped a production document-extraction system with Claude Opus 4.7 and Amazon Textract on AWS, integrated with LIMS via REST API, processing 1,800 lab documents and automating 800 Jira tickets.

Senior Machine Learning Engineer | Viavi Solutions | Chennai, India Sep 2024 – Nov 2025

- Built an event-correlation engine clustering alarms with Sentence-BERT bi-encoder embeddings and DBSCAN to trace cascading network alarm floods back to the root event, improving semantic classification by 73.5%.
- Built an internal Project Discovery Assistant combining vector search over embeddings with LLM re-ranking and reasoning, reaching a 90% match rate against existing organizational solutions.
- Drove technical direction on 15+ ML applications from ambiguous requirements, owning problem discovery, feature and model choice, offline evaluation and delivery, with 7 reaching production or paid client work.

Data Scientist | Viavi Solutions | Bengaluru, India Nov 2021 – Sep 2024

- Deployed a production SLA breach prediction system with LightGBM gradient boosting, FastAPI, Docker and MLflow, using GPT-3.5 for real-time root-cause analysis per breach, reducing downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, retrained on Airflow schedules and monitored in production for drift.
- Delivered technical POCs and consulting for 15+ clients across 3+ ML platforms, aligning stakeholders on scope and success metrics and contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | Optum | Gurugram, India Jun 2021 – Nov 2021

- Implemented a re-ranker over the candidate set of a production recommendation system serving individual and child plan recommendations, engineering ranking features that balanced customer relevance against renewal targets and lifting conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

Data Engineer | Nippon Data Systems | Gurugram, India Nov 2019 – May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL with multi-table joins, CTEs and window functions, delivering ad-hoc analyses for multiple stakeholder teams.

Associate Consultant | Saksoft | Gurugram, India Dec 2017 – Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

PROJECTS

Think Tree – ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Multi-Objective Home-Feed Ranking Prototype

- Built a personalised home-feed ranker over a marketplace catalogue: two-tower embedding candidate generation with vector search, a LightGBM learning-to-rank re-ranker, and multi-objective blending of customer relevance against commercial guardrails, with cold-start exploration and offline NDCG and recall evaluation to decide what would be worth testing online.

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection – Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 – Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 – Aug 2017

TECHNICAL SKILLS

Ranking / Recommenders: Learning-to-Rank, Recommender Systems, Personalisation, Re-Ranking, Candidate Generation, Collaborative Filtering, Two-Tower Models, Cross-Encoder, Embeddings, Vector Search, Cold-Start, Multi-Objective Optimisation, Information Retrieval, Apache Solr, Lucene Indexing

Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Gradient Boosting, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Deep Learning

Experimentation / Evaluation: Offline and Online Evaluation, A/B Testing, Experiment Design, Statistical Inference, Statistical Modeling, NDCG, Precision and Recall, Model Monitoring, Drift Detection, LLM Evals, Tracing

Programming / Data: Python, SQL, PySpark, Apache Spark, Pandas, NumPy, Spark SQL, Hive, FastAPI, Flask, REST API, Pydantic, Unix

MLOps / Cloud: AWS, SageMaker, Amazon Textract, Docker, Kubernetes, Airflow, MLflow, DVC, Terraform, Jenkins, Bitbucket, GCP, Azure

AI / GenAI: LLM, RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Fine-Tuning, Hugging Face Transformers, NLP

AI Coding Tools: Claude Code, Cursor

Databases: PostgreSQL, TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies